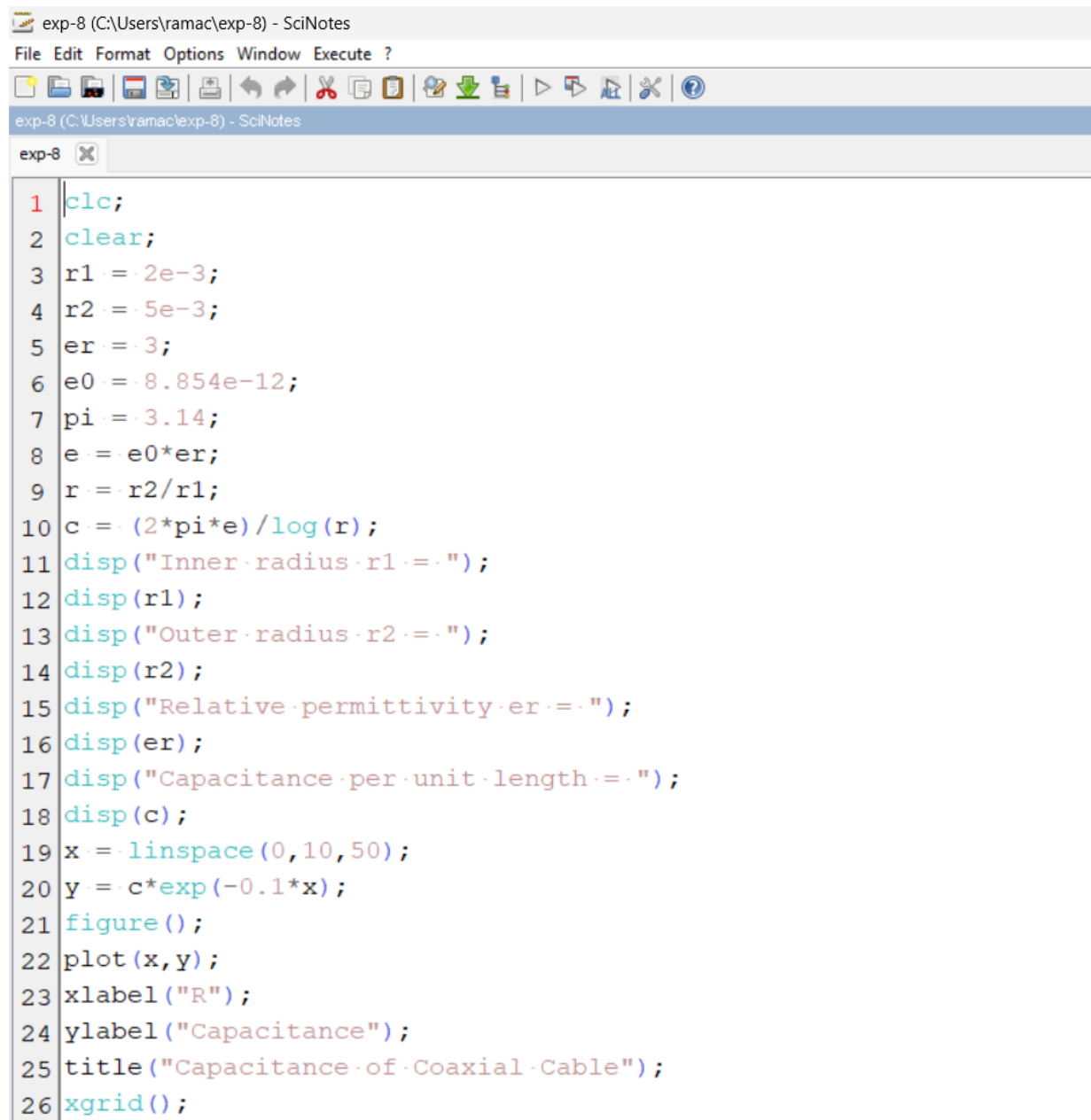


Exp:8 -Determination and plotting of Capacitance of Co-axial cable per unit length with different (i)inner radius, (ii) outer radius and (iii) relative permittivity of the dielectric medium.

Aim: To write program for Capacitance of co-axial cable per unit length using SCILAB.

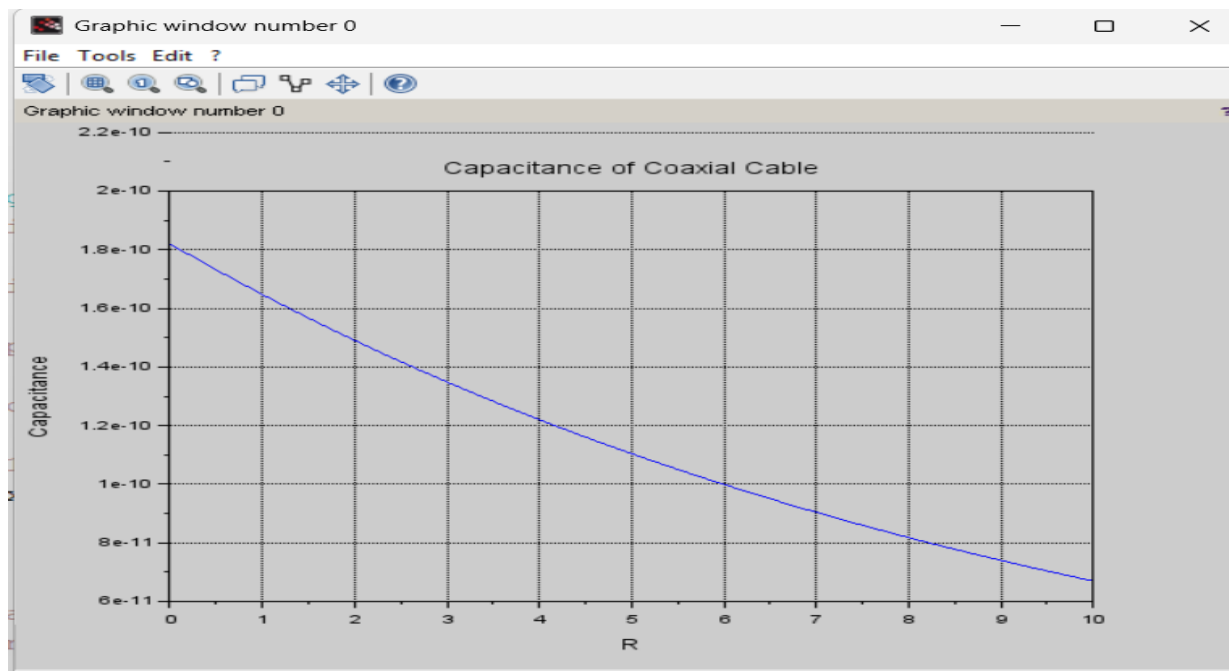
Software required: SCILAB version 6.0.1 **Practical**

output:



```
exp-8 (C:\Users\ramac\exp-8) - SciNotes
File Edit Format Options Window Execute ?
exp-8 (C:\Users\ramac\exp-8) - SciNotes
exp-8

1 clc;
2 clear;
3 r1 = 2e-3;
4 r2 = 5e-3;
5 er = 3;
6 e0 = 8.854e-12;
7 pi = 3.14;
8 e = e0*er;
9 r = r2/r1;
10 c = (2*pi*e)/log(r);
11 disp("Inner radius r1 = ");
12 disp(r1);
13 disp("Outer radius r2 = ");
14 disp(r2);
15 disp("Relative permittivity er = ");
16 disp(er);
17 disp("Capacitance per unit length = ");
18 disp(c);
19 x = linspace(0,10,50);
20 y = c*exp(-0.1*x);
21 figure();
22 plot(x,y);
23 xlabel("R");
24 ylabel("Capacitance");
25 title("Capacitance of Coaxial Cable");
26 xgrid();
```



Scilab 6.1.1 Console

File Edit Control Applications ?

Scilab 6.1.1 Console

```
"Inner radius r1 = "  
0.002  
"Outer radius r2 = "  
0.005  
"Relative permittivity er = "  
3.  
"Capacitance per unit length = "  
1.820D-10  
--> |
```

Theoretical Output:

permittivity

$$\epsilon = \epsilon_0 \epsilon_r$$

$$= (8.854 \times 10^{-12}) (3)$$

$$\Rightarrow 2.6562 \times 10^{-11} \text{ F/m}$$

$$\text{radius} \Rightarrow \frac{r_2}{r_1} \Rightarrow \frac{5 \times 10^{-3}}{2 \times 10^{-3}}$$

$$\Rightarrow 2.5$$

$$\left(\frac{r_2}{r_1} \right) \Rightarrow \ln(2.5)$$

$$\Rightarrow 0.9163$$

capacitance $\Rightarrow C = \frac{2\pi\epsilon}{\ln(r_2/r_1)}$

$$C = \frac{2(3.14)(2.6562 \times 10^{-11})}{0.9163}$$

$$C = \frac{1.668 \times 10^{-10}}{0.9163}$$

$$C \approx 1.82 \times 10^{-10} \text{ F/m.}$$

Result :

Thus the program was verified for Capacitance of co-axial cable per unit length using SCILAB was successfully.